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### Academic Curriculum

Associate Professor of Computer Science (INF/01) in the Department of Communication Sciences, he obtained the qualification as Full Professor in ING-INF/05 (Information Processing Systems). His research fields include coding theory and artificial intelligence. He also deals with the implementation of signal processing and pattern recognition algorithms, fuzzy logic, neural networks and genetic algorithms. Fixed-term researcher art. 24 c.3-b L. 240/10 (2016-2019) and L. 230/2005 (2009-2012) in Computer Science at the University of Teramo, has carried out research in the field of optimization of control systems and database project at the Astronomical Observatory of Collurania "V. Cerulli" in Teramo. Volume editor Springer and Elsevier, Associate Editor of the journals IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Emerging Topics in Computational Intelligence, IEEE Access, Array (Elsevier), International Journal of Machine Learning and Cybernetics (Springer), is a member of Machine Intelligence Research Labs (MIR Labs). Honorary Professor of Ramco Institute of Technology, Rajapalayam (Anna University) and Rathinam Technical Campus, Coimbatore (Anna University), Tamilnadu, India, he is Adjunct Professor in Computer Science at School of Engineering and Technology of Shobhit University, Gangoh, India. Honorary Professor at the Department of Computer Science & Engineering of the University of Galgotias, Greater Noida, Uttar Pradesh, India. Visiting and Honorary Professor at School of Computing, Faculty of Engineering, Bharath Institute of Higher Education and Research, Chennai, Tamil Nadu, India. World's 2% Top Scientist 2021, is Keynote Speaker and Guest of Honor of IEEE conferences and Guest Editor of Special Issue of journals Elsevier, Springer and Inderscience, is Chair of IEEE conferences and member of the editorial board of various international journals and of the scientific committee of international conferences. Author of the invention of patents on Artificial Intelligence, member of the Doctoral Board in Economic and Social Sciences (University of Teramo), he is PhD in Computational Astrophysics. He holds a degree in Physics from the University of Bologna. he is Keynote Speaker and Guest of Honor of IEEE conferences and Guest Editor of Special Issues of Elsevier, Springer and Inderscience journals, he is Chair of IEEE conferences and member of the editorial board of various international journals and of the scientific committee of international conferences. Author of the invention of patents on Artificial Intelligence, member of the Doctoral Board in Economic and Social Sciences (University of Teramo), he is PhD in Computational Astrophysics. He holds a degree in Physics from the University of Bologna. he is Keynote Speaker and Guest of Honor of IEEE conferences and Guest Editor of Special Issues of Elsevier, Springer and Inderscience journals, he is Chair of IEEE conferences and member of the editorial board of various international journals and of the scientific committee of international conferences. Author of the invention

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## Publications

- Vimal, S., Rho, S., Pelusi, D. Special Issue on Artificial Intelligence Empowered Big Data Analytical Patterns for Medical Applications. *Neural Process Lett* (2023). <https://doi.org/10.1007/s11063-023-11158-8>
- Jude Hemanth, Danilo Pelusi, Joy long-Zong Chen, Intelligent Cyber Physical Systems and Internet of Things - ICoICI 2022, Engineering Cyber-Physical Systems and Critical Infrastructures, Vol. 3, Springer, February 2023.
- Aparna Pramanik, Asit Kumar Das, Danilo Pelusi, Janmenjoy Nayak, An Effective Fuzzy Clustering of Crime Reports Embedded by a Universal Sentence Encoder Model, *Mathematics*, 2023, 11(3), 611.
- H. Swapnarekha, Janmenjoy Nayak, HS Behera, Pandit Byomakesha Dash, Danilo Pelusi, An optimistic firefly algorithm-based deep learning approach for sentiment analysis of COVID-19 tweets, *Mathematical Biosciences and Engineering*, 2023, Volume 20, Issue 2: 2382- 2407
- Oram, E., Naik, B., Bhoi, G., Pelusi, D. (2023). Heart Failure Detection from Clinical and Lifestyle Information using Optimized XGBoost with Gravitational Search Algorithm. In: Nayak, J., Das, AK, Naik, B., Meher, SK, Brahman, S. (eds) *Nature-Inspired Optimization Methodologies in Biomedical and Healthcare*. Intelligent Systems Reference Library, vol 233. Springer, Cham. [https://doi.org/10.1007/978-3-031-17544-2\\_5](https://doi.org/10.1007/978-3-031-17544-2_5)
- Bhoi, G., Oram, E., Naik, B., Pelusi, D. (2023). Optimized Gradient Boosting Tree-Based Model for Obesity Level Prediction from patient's Physical Condition and Eating Habits. In: Nayak, J., Das, AK, Naik, B., Meher, SK, Brahman, S. (eds) *Nature-Inspired Optimization Methodologies in Biomedical and Healthcare*. Intelligent Systems Reference Library, vol 233. Springer, Cham. [https://doi.org/10.1007/978-3-031-17544-2\\_3](https://doi.org/10.1007/978-3-031-17544-2_3)
- Pemmada, SK, Nayak, J., Behera, HS, Pelusi, D. (2022). Light Gradient Boosting Machine in Software Defect Prediction: Concurrent Feature Selection and Hyper Parameter Tuning. In: Raj, JS, Shi, Y., Pelusi, D., Balas, VE (eds) *Intelligent Sustainable Systems*. Lecture Notes in Networks and Systems, vol 458. Springer, Singapore. [https://doi.org/10.1007/978-981-19-2894-9\\_32](https://doi.org/10.1007/978-981-19-2894-9_32)
- Jennifer S. Raj, Yong Shi, Danilo Pelusi, Valentina Emilia Balas, Intelligent Sustainable Systems - Proceedings of ICISS 2022, Lecture Notes in Networks and Systems, vol 458. Springer, Singapore. [https://doi.org/10.1007/978-981-19-2894-9\\_2](https://doi.org/10.1007/978-981-19-2894-9_2)
- Ramani, S., Ghiya, A., Aravind, PS, Karuppiiah, M., Pelusi, D. (2022). Predicting New York Taxi Trip Duration Based on Regression Analysis Using ML and Time Series Forecasting Using DL. In: Raj, JS, Shi, Y., Pelusi, D., Balas, VE (eds) *Intelligent Sustainable Systems*. Lecture Notes in Networks and Systems, vol 458. Springer, Singapore. [https://doi.org/10.1007/978-981-19-2894-9\\_2](https://doi.org/10.1007/978-981-19-2894-9_2)

- Janmenjoy Nayak, Manohar Mishra, Danilo Pelusi, Bighnaraj Naik, Chapter Eight - Internet of things and fog computing application to improve the smart-grid resiliency, Electric Power Systems Resiliency, Academic Press, 2022, Pages 213-229, ISBN 9780323855365, doi: 10.1016/B978-0-323-85536-5.00005-9.
- Manohar Mishra, Renu Sharma, Danilo Pelusi, Bighnaraj Naik, Editorial, Special Issue on: Smart Computational Intelligence and Optimization Method Applications to Electrical Power Engineering, International Journal of Innovative Computing and Applications, 2022, Vol.13, No.2.
- Nayak, J., Al-Dabass, D., Pelusi, D., Manohar, M., Special issue on machine learning for security and privacy: advancing the state-of-the-art applications. Neural Computing and Applications (2022). <https://doi.org/10.1007/s00521-022-08183-4>
- Asit Kumar Das, Janmenjoy Nayak, Bighnaraj Naik, S. Vimal, Danilo Pelusi, Computational Intelligence in Pattern Recognition - Proceedings of CIPR 2022, Lecture Notes in Networks and Systems, Springer, XXVIII, 665, ISBN:978-981-19-3088 -1, 2022.
- Xiaozhuan Gao; Lipeng Pan; Danilo Pelusi; Yong Deng, Fuzzy Markov Decision-Making Model for Interference Effects, IEEE Transactions on Fuzzy Systems, 21 June 2022, DOI: 10.1109/TFUZZ.2022.3184784.
- Janmenjoy Nayak, HS Behera, Bighnaraj Naik, S. Vimal, Danilo Pelusi, Computational Intelligence in Data Mining - Proceedings of ICCIDM 2021, Smart Innovation, Systems and Technologies, Springer, ISBN: 978-981-16-9446-2, 2022.
- D. Jude Hemanth, Danilo Pelusi, Chandrasekar Vuppalapati, Intelligent Data Communication Technologies and Internet of Things - Proceedings of ICICI 2021, Lecture Notes on Data Engineering and Communications Technologies, Springer, ISBN: 978-981-16-7609-3, 2022
- S. Gopikumar, J. Rajesh Banu, Y. Harold Robinson, S. Raja, S. Vimal, Danilo Pelusi & M Kaliappan (2022) Geo Spatial Based Real Time Monitoring on Eutrophic Evaluation of Porunai River Basin for Pollution Risk Assessment, European Journal of Remote Sensing, DOI: 10.1080/22797254.2021.2025152
- Sankhadeep Chatterjee, Asit Kumar Das, Janmenjoy Nayak, Danilo Pelusi, Improving Facial Emotion Recognition Using Residual Autoencoder Coupled Affinity Based Overlapping Reduction, Mathematics, 10(3), 406, 2022.
- S. Thanga Revathi, A. Gayathri, J. Kalaivani, Mary Subaja Christo, Danilo Pelusi, M. Azees, Cloud-Assisted Privacy-Preserving Method for Healthcare Using Adaptive Fractional Brain Storm Integrated Whale Optimization Algorithm, Security and Communication Networks, vol. 2021, Article ID 6210054, 10 pages, 2021.
- S. Satheeskumaran, Zhang Yu-Dong, Danilo Pelusi, Guest editorial, Intelligent edge and 5G technologies for pervasive computing and communications, International Journal of Pervasive Computing and Communications, Vol. 17 No. 5, pp. 445-446. <https://doi.org/10.1108/IJPCC-11-2021-216>
- Jamenjoy Nayak, Danilo Pelusi, Bighnaraj Naik, Manohar Mishra, Khan Muhammad, David Al-Dabass, Computational Intelligence in Cancer Diagnosis: Progress and Challenges, 1st Edition, Academic Press, ISBN: 9780323852401, June 1, 2022.
- Asit Kumar Das, Janmenjoy Nayak, Bighnaraj Naik, Soumi Dutta, Danilo Pelusi, Computational Intelligence in Pattern Recognition - Proceedings of CIPR 2021, Advances in Intelligent Systems and Computing, Vol. 1349, Springer, ISBN 978-981-16-2542-8 , 2022.

- Yangxue Li, Danilo Pelusi, Yong Deng, Kang Hao Cheong, Relative Entropy of Z-numbers, Information Sciences, August 2021, <https://doi.org/10.1016/j.ins.2021.08.077>
- Deepaa Selva, Balakrishnan Nagaraj, Danilo Pelusi, Rajendran Arunkumar, Ajay Nair, Intelligent Network Intrusion Prevention Feature Collection and Classification Algorithms, Algorithms, 14(8), 224, 2021, <https://doi.org/10.3390/a14080224>
- Li, Y., Pelusi, D., Cheong, KH, Deng Y., The arithmetics of two dimensional belief functions. Applied Intelligence (2021). <https://doi.org/10.1007/s10489-021-02435-x>
- Pavithra Latha Kumaresan, Subbulakshmi Pasupathi, Sindhia Lingaswamy, Sreesharmila Thangaswamy, Vimal Shunmuganathan, Danilo Pelusi, Fruit-Fly optimization based feature integration in image retrieval, Mathematical Biosciences and Engineering, 2021, Volume 18, Issue 5: 6178-6197. doi: 10.3934/mbe.2021309
- Meenu Garg, Sheifali Gupta, Soumya Ranjan Nayak, Janmenjoy Nayak, Danilo Pelusi, Modified pixel level snake using bottom hat transformation for evolution of retinal vasculature map, Mathematical Biosciences and Engineering, Volume 18, Issue 5: 5737-5757, 2021, doi: 10.3934/mbe.2021290.
- Weiping Ding, Janmenjoy Nayak, H.Swapnarekha, Ajith Abraham, Bighnaraj Naik, Danilo Pelusi, Fusion of Intelligent Learning for COVID-19: A State-of-the-Art Review and Analysis on Real Medical Data, Neurocomputing, 2021, <https://doi.org/10.1016/j.neucom.2021.06.024>
- Sakshi Patel, V. Thanikaiselvan, Danilo Pelusi, B. Nagaraj, R. Arunkumar, R. Amirtharajan, Color image encryption based on customized neural network and DNA encoding, Neural Computing and Applications (2021). <https://doi.org/10.1007/s00521-021-06096-2>
- Janmenjoy Nayak, P. Suresh Kumar, Dukka Karun Kumar Reddy, Bighnaraj Naik, Danilo Pelusi, Machine Learning and Big Data in Cyber-Physical System: Methods, Applications and Challenges, Cognitive Engineering for Next Generation Computing: A Practical Analytical Approach, pp. 49-91, 2021, <https://doi.org/10.1002/9781119711308.ch2>
- Janmenjoy Nayak, Bighnaraj Naik, Paidi Dinesh, Kanithi Vakula, Pandit Byomakesha Dash, Danilo Pelusi, Significance of deep learning for Covid-19: state-of-the-art review, Research on Biomedical Engineering (2021), <https://doi.org/10.1007/s42600-021-00135-6>
- Janmenjoy Nayak, Bighnaraj Naik, Pandit Byomakesha Dash, Danilo Pelusi, Optimal Fuzzy Cluster Partitioning by Crow Search Meta-Heuristic for Biomedical Data Analysis, International Journal of Applied Metaheuristic Computing (IJAMC) 12(2), 2021
- Janmenjoy Nayak, Bighnaraj Naik, Danilo Pelusi, Asit Das, Handbook of Computational Intelligence in Biomedical Engineering and Healthcare 1st Edition, Academic Press, April 2021.
- Satheeskumaran, S., Zhang, Y.-D., Pelusi, D. (2020), "Guest editorial", International Journal of Intelligent Unmanned Systems, Vol. 9 No. 1, pp. I-II. <https://doi.org/10.1108/IJIUS-01-2021-078>
- Rekha HS, Nayak J., Sekhar GTC, Pelusi D., Impact of IoT in Healthcare: Improvements and Challenges. In: Das K., Mishra BSP, Das M. (eds) The Digitalization Conundrum in India. India Studies in Business and Economics (2020). Springer, Singapore. [https://doi.org/10.1007/978-981-15-6907-4\\_5](https://doi.org/10.1007/978-981-15-6907-4_5)
- Janmenjoy Nayak, Bighnaraj Naik, Asit Kumar Das, Danilo Pelusi, Nature inspired optimization and its application to engineering, Evolutionary Intelligence, Springer (2021)

- Chirantana Mallick, Asit Kumar Das, Janmenjoy Nayak, Danilo Pelusi, S. Vimal, Evolutionary Algorithm based Ensemble Extractive Summarization for Developing Smart Medical System, *Interdisciplinary Sciences: Computational Life Sciences* (2021). <https://doi.org/10.1007/s12539-020-00412-5>
- Priyanka Das, Asit Kumar Das, Janmenjoy Nayak, Danilo Pelusi, Weiping Ding, Incremental Classifier in Crime Prediction using bi-objective Particle Swarm Optimization, *Information Sciences*, 2021, <https://doi.org/10.1016/j.ins.2021.02.002>
- Janmenjoy Nayak, Bighnaraj Naik, Paidi Dinesh, Kanithi Vakula, B. Kameswara Rao, Weiping Ding, Danilo Pelusi, Intelligent system for COVID-19 prognosis: a state-of-the-art survey. *Applied Intelligence* (2021). <https://doi.org/10.1007/s10489-020-02102-7>
- Sekhar, GTC, Behera, HS, Nayak, J., Naik, B., Pelusi, D., Intelligent Computing in Control and Communication - Proceeding of the First International Conference on Intelligent Computing in Control and Communication (ICCC 2020), LECTURE NOTES IN ELECTRICAL ENGINEERING - ISSN:1876-1100 vol. 702, Springer, 2021.
- Balakrishnan, Nagaraj; Pelusi, Danilo; Ganesan, Subramaniam, Special issue on “Big Data Analytics and Deep Learning for E-Business Outcomes”. DOI:10.1007/s10257-020-00486-0. pp.281-282. In *Information Systems and e-Business Management* - ISSN:1617-9846 vol. 18(3), 2020.
- Sharma, R., Mishra, M., Nayak, J., Naik, B., Pelusi, D., Green Technology for Smart City and Society - Proceedings of GTSCS 2020, Lecture Notes in Networks and Systems, Springer, ISSN:2367 -3370 vol. 151, 2021
- Li, Y., Pelusi, D. & Deng, Y. Generate two-dimensional belief function based on an improved similarity measure of trapezoidal fuzzy numbers. *Comp. Appl. math.* **39**, 326 (2020). <https://doi.org/10.1007/s40314-020-01371-9>
- Praveen Kumar Malik, Rohit Sharma, Rajesh Singh, Anita Gehlot, Suresh Chandra Satapathy, Waleed S. Alnumay, Danilo Pelusi, Uttam Ghosh, Janmenjoy Nayak, Industrial Internet of Things and its applications in industry 4.0: State of the art, *Computer Communications*, 2020, <https://doi.org/10.1016/j.comcom.2020.11.016>.
- Nagaraj, B., Pelusi, D., Chen JI-Z., Special section on emerging challenges in computational intelligence for signal processing applications, *Intelligent Automation and Soft Computing*, Volume 26, Issue 4, 2020, Pages 737-739.
- D. Pelusi, R. Mascella and L. Tallini, "A fitness dependent salp swarm algorithm," 2020 IEEE Congress on Evolutionary Computation (CEC), Glasgow, United Kingdom, 2020, pp. 1-6, doi: 10.1109/CEC48606.2020.9185594.
- Janmenjoy Nayak, Kanithi Vakula, Paidi Dinesh, Bighnaraj Naik, Danilo Pelusi, "Intelligent food processing: Journey from artificial neural network to deep learning", *Computer Science Review*, Volume 38, November 2020.
- W. Ding, J. NAYAK, B. Naik, D. Pelusi and M. Mishra, "Fuzzy and Real Coded Chemical Reaction Optimization for Intrusion Detection in Industrial Big Data Environment," in *IEEE Transactions on Industrial Informatics*, doi: 10.1109/TII.2020.3007419.
- Ghazaala Yasmin, Asit Kumar Das, Janmenjoy Nayak, Danilo Pelusi, Weiping Ding, Graph based Feature Selection Investigating Boundary Region of Rough Set for

Language Identification, Expert Systems with Applications, DOI:10.1016/j.eswa.2020.113575, pp.113575, 2020 .

- Nayak, J., Chandrasekhar, GT, Naik, B. , Danilo Pelusi and A. Abraham, Special issue on "Soft computing techniques: applications and challenges" neural computing and applications. *Neural Computing & Applications* (2020). <https://doi.org/10.1007/s00521-020-04902-x>.
- Tao Wen, Danilo Pelusi, Yong Deng, Vital spreaders identification in complex networks with multi-local dimension, *Knowledge-Based Systems*, DOI:10.1016/j.knosys.2020.105717, 2020.
- Shengzhong Mao, Yong Deng, Danilo Pelusi, Alternatives selection for produced water management: A network-based methodology, *Engineering Applications of Artificial Intelligence* Volume 91, May 2020.
- Shengzhong Mao, Yuzhen Han, Yong Deng, Danilo Pelusi, A hybrid DEMATEL-FRACTAL method of handling dependent evidences, *Engineering Applications of Artificial Intelligence*, Volume 91, May 2020.
- Balakrishnan Nagaraj, Danilo Pelusi, Editorial, *International Journal of Biometrics*, Volume 12, Issue 1, 2020, Pages 1-2
- Renu Sharma, Manohar Mishra, Janmenjoy Nayak, Bighnaraj Nayak, Danilo Pelusi, *Innovation in Electrical Power Engineering, Communication, and Computing Technology - Proceedings of IEPCCT 2019, Lecture Notes in Electrical Engineering*, Springer, Vol. 630, 2020.
- Asit Kumar Das, Janmenjoy Nayak, Bighnaraj Naik, Soumi Dutta, Danilo Pelusi, *Computational Intelligence in Pattern Recognition - Proceedings of CIPR 2020, Advances in Intelligent Systems and Computing*, vol. 1120, Springer, 2020
- B.Naik, M. Obaidat, J. Nayak, D. Pelusi, P. Vijayakumar, Sk Hafizul Islam, "Intelligent Secure Ecosystem based on Meta-heuristic and Functional Link Neural Network for Edge-of-Things", *IEEE Transactions on Industrial Informatics*, Vol 16(3), 1947-1956, 2020.
- Nayak J., Naik B., Pelusi D., Krishna AV (2020) A Comprehensive Review and Performance Analysis of Firefly Algorithm for Artificial Neural Networks. In: Yang XS., He XS. (eds) *Nature-Inspired Computation in Data Mining and Machine Learning. Studies in Computational Intelligence*, vol 855. Springer, Cham
- Behera, HS, Nayak, J., Naik, B., Pelusi, D., Preface, *Advances in Intelligent Systems and Computing*, Volume 990, 2020, Page xv.
- Renu Sharma, Manohar Mishra, Janmenjoy Nayak, Bighnaraj Naik, Danilo Pelusi (2020), *Innovation in Electrical Power Engineering, Communication, and Computing Technology: Proceedings of IEPCCT 2019, Lecture Notes in Electrical Engineering*, Springer, Vol. 630.
- Danilo Pelusi, Raffaele Mascella, Luca Tallini, Janmenjoy Nayak, Bighnaraj Naik, Yong Deng, Improving exploration and exploitation via a Hyperbolic Gravitational Search Algorithm, *Knowledge-Based Systems* (2019), DOI: 10.1016/j.knosys.2019.105404
- Danilo Pelusi, Raffaele Mascella, Luca Tallini, Janmenjoy Nayak, Bighnaraj Naik, Yong Deng. An Improved Moth-Flame Optimization algorithm with hybrid search phase, *Knowledge-Based Systems*, 2019, DOI: 10.1016/j.knosys.2019.105277

- Priyanka Das, Asit Kumar Das, Janmenjoy Nayak, Danilo Pelusi, Weiping Ding. Group Incremental Adaptive Clustering based on Neural Network and Rough Set Theory for Crime Report Categorization, *Nerocomputing*, 2019, DOI: 10.1016/j.neucom.2019.10.109
- Swapna Rekha H., Nayak J., Naik B., Pelusi D. (2020) Soft Computing in Robotics: A Decade Perspective. In: Nayak J., Balas V., Favorskaya M., Choudhury B., Rao S., Naik B. (eds) *Applications of Robotics in Industry Using Advanced Mechanisms. ARIAM 2019. Learning and Analytics in Intelligent Systems*, vol 5. Springer, Cham
- Balakrishnan, Nagaraj; Rajendran, Arunkumar; Pelusi, Danilo; Ponnusamy, Vijayakumar. (2019). Deep Belief Network enhanced Intrusion Detection System to Prevent Security Breach in the Internet of Things, *Internet of Things*, DOI:10.1016/j.iot.2019.100112. pp.100112.
- Yurui Ming, Weiping Ding, Danilo Pelusi, Dongrui Wu, Yu-Kai Wang, Mukesh Prasad, Chin-Teng Lin, Subject adaptation network for EEG data analysis, *Applied Soft Computing*, 2019, <https://doi.org/10.1016/j.asoc.2019.105689>
- Danilo Pelusi, On Redundancy Reduction of Non-Recursive Second-Order Spectral-Null Codes, *IEEE Access*, Vol. 7, pp. 112171-112179, 2019.
- Janmenjoy Nayak, Paidi Dinesh, Kanithi Vakula.Bighnaraj Naik, Danilo Pelusi, Applications and Advancements of Nature-Inspired Optimization Algorithms in Data Clustering: A Detailed Analysis, *Computational Intelligence in Data Mining, Advances in Intelligent Systems and Computing*, Springer, pp 731- 750, 2019.
- Himansu Sekhar Behera, Janmenjoy Nayak, Bighnaraj Naik, Danilo Pelusi, *Computational Intelligence in Data Mining Proceedings of the International Conference on ICCIDM 2018*, Springer 2019.
- Asit Kumar Das, Janmenjoy Nayak, Bighnaraj Naik, Soumen Kumar Pati, Danilo Pelusi, *Computational Intelligence in Pattern Recognition, Proceedings of CIPR 2019*, Springer, 2019.
- Priyanka Das, Asit Kumar Das, Janmenjoy Nayak, Danilo Pelusi, Weiping Ding, "A Graph Based Clustering Approach for Relation Extraction From Crime Data", *IEEE Access*, Vol. 7, 101269-101282, 2019.
- P. Das, AK Das, J. Nayak, D. Pelusi, "A framework for crime data analysis using relationship among named entities", *Neural Computing and Applications*, 2019, <https://doi.org/10.1007/s00521-019-04150-8>
- Balakrishnan, N., Pelusi, D., Chen, JI-Z., "Editorial", *International Journal of Business Intelligence and Data Mining*, Volume 14, Issue 1-2, 2019, Pages 1-2.
- Yurui Ming, Danilo Pelusi, Chieh-Ning Fang, Mukesh Prasad, Yu-Kai Wang, Dongrui Wu, Chin-Teng Lin, "EEG data analysis with stacked differentiable neural computers", *Neural Computing and Applications*, 2018, <https://doi.org/10.1007/s00521-018-3879-1>.
- Gorripotu, TS, Samalla, H., Jagan Mohana Rao, C., Azar, AT, Pelusi, D., "TLBO algorithm optimized fractional-order PID controller for AGC of interconnected power system", *Advances in Intelligent Systems and Computing*, Volume 758, pp. 847-855, 2018.

- Nayak, J., Paparao, S., Naik, B., Seetayya, N., Pradeep, P., Behera, HS, Pelusi, D., "Chemical reaction optimization: A survey with application and challenges", *Advances in Intelligent Systems and Computing*, Volume 758, pp. 507-524, 2018.
- Pelusi D., Mascella R., Tallini L. (2018), "A Fuzzy Gravitational Search Algorithm to Design Optimal IIR Filters", *Energies*, vol. 11(4), 736.
- Naik B., Mishra D., Nayak J., Pelusi D., Abraham A., "Perturbation Based Efficient Crow Search Optimized FLANN for System Identification: A Novel Approach", *Hybrid Intelligent Systems (HIS) 2017*, *Advances in Intelligent Systems and Computing*, Vol. 734, p. 204-212.
- Pelusi D., Mascella R., Tallini L., Nayak J., Naik B., Abraham A., "Neural Network and Fuzzy System for the tuning of Gravitational Search Algorithm parameters", *Expert Systems with Applications*, Vol. 102, 234-244, 2018.
- Pelusi, D.; Maxilla, R.; Tallini, L., "Revised Gravitational Search Algorithms Based on Evolutionary-Fuzzy Systems", *Algorithms*, Vol. 10, Issue 2, 44, 2017.
- Nagaraj, B., Pelusi, D., Che, Ji-Z., Introductory Editorial, *Wireless Personal Communications*, pp. 1-2, 2017
- D. Pelusi, R. Mascella, LG Tallini, L. Vazquez, D. Diaz, Control of Drum Boiler Dynamics via an Optimized Fuzzy Controller, *International Journal of Simulation Systems, Science & Technology*, Volume 17, Number 33, pp. 22.1-22.7, 2016
- LG Tallini, D. Pelusi, R. Mascella, L. Pezza, S. Elmougy, B. Bose, Efficient Non-Recursive Design of Second-Order Spectral-Null Codes. *IEEE Transactions on Information Theory*, Vol. 62, Issue 6, pp. 3084-3102, 2016.
- D. Pelusi, M. Tivegna, Explorations in the use of artificial intelligence techniques and econometric forecasting in the Euro-\$ market, *International Conference on Mathematical and Statistical Methods for actuarial sciences and finance*, Paris, France, March 30-31, April 1, 2016
- G. Taffoni, E. Sciacca, A. Pietrinferni, U. Becciani, A. Costa, S. Cassisi, F. Pasian, D. Pelusi, C. Vuerli, Feeding an astrophysical database via distributed computing resources: The case of BaSTI, *Astronomy and Computing*, Volume 11, Part B, Pages 109-118, June 2015,
- Pelusi D, Elmougy S, Tallini LG, Bose B., m-ary Balanced Codes with Parallel Decoding, *IEEE Transactions on Information Theory*, Vol. 61, Issue 6, pp. 3251-3264, 2015
- A. Pietrinferni, M. Molinaro, S. Cassisi, F. Pasian, M. Salaris, D. Pelusi, P. Manzato, C. Vuerli, BaSTI: An updated, advanced and VO-compliant database of stellar evolution predictions, *Astronomy and Computing*, Volumes 7-8, pp. 95-100, November-December 2014
- M. Tivegna, P. Ippoliti, D. Pelusi, Intelligent algorithms for trading the Euro-Dollar with a news-based econometric model of the foreign exchange market, *International Conference on Mathematical and Statistical Methods for actuarial sciences and finance*, Vietri sul Mare, Italy, April 22-24, 2014
- Pelusi D, Tivegna M, Ippoliti P, Intelligent algorithms for trading the Euro-Dollar in the foreign exchange market. *Mathematical and Statistical Methods for Actuarial Sciences and Finance*, pp. 243-252, ISBN 978-3-319-02499-8, Springer, 2014



- R. Mascella, D. Pelusi, L. Pezza, S. Elmougy, LG Tallini, B. Bose, "On Efficient Second-Order Spectral-Null Codes using Sets of m1-Balancing Functions", IEEE International Symposium on Information Theory, pp . 141-145, July 7-12, 2013
- Pelusi D, Designing neural networks to improve timing performances of intelligent controllers. Journal of Discrete Mathematical Science & Cryptography, Vol. 16(2-3), pp. 187-193, 2013
- Pelusi D, Tivegna M, Ippoliti P, Improving the profitability of Technical Analysis through intelligent algorithms. Journal of Interdisciplinary Mathematics, Vol. 16(2-3), pp. 203-215, 2013
- D. Pelusi, L. Vazquez, D. Diaz, R. Mascella, "Fuzzy Algorithm Control Effectiveness on Drum Boiler Simulated Dynamics", International Conference on Telecommunications and Signal Processing (TSP), ISBN 978-1-4799-0403-7, ISSN 1805-5435, pp. 272-276, 2013
- D. Pelusi, R. Mascella, "Optimal control algorithms for second order systems", Journal of Computer Science, Science Publications, Vol. 9, Issue 2, pp. 183-197, 2013
- D. Pelusi, "PID and Intelligent Controllers for optimal timing performances of industrial actuators", International Journal of Simulation, Systems, Science & Technology, ISSN 1473-804x Online, ISSN 1473-8031 Print, Vol. 13, No. 2, pp . 65-71, 2012.
- D. Pelusi, M. Tivegna, P. Ippoliti, "Intelligent algorithms for trading the Euro-Dollar in the foreign exchange market", proceedings of the International Conference on Mathematical and Statistical Methods for Actuarial Sciences and Finance, Venice, Italy, 2012.
- D. Pelusi, "Improving Settling and Rise Times of Controllers Via Intelligent Algorithms", proceedings of the IEEE International Conference on Modeling and Simulation, ISBN: 978-0-7695-4682-7/12, pp. 187-192, Cambridge, United Kingdom, 2012
- D. Pelusi, M. Tivegna, "Optimal trading rules at hourly frequency in the foreign exchange markets", Mathematical and Statistical Methods for Actuarial Sciences and Finance, Springer, Hardcover ISBN: 978-88-470-2341-3, pp. 341-348, 2012
- D. Pelusi, "On designing optimal control systems through genetic and neuro-fuzzy techniques", proceedings of the IEEE International Symposium on Signal Processing and Information Technology, ISBN: 978-1-4673-0753-6, pp. 134-139, Bilbao, Spain, 2011.
- D. Pelusi, "Design and implementation of fuzzy controllers for optimal signals sampling", International Conference on Instrumentation, Measurement, Circuits and Systems (ICIMCS 2011), ASME, ISBN: 978-0-7918-59902, Volume 2, pp. 485-488, 2011.
- D. Pelusi, "Genetic-Neuro-Fuzzy Controllers for second order control systems", proceedings of the IEEE European Modeling Symposium on Mathematical Modeling and Computer Simulation, ISBN: 978-1-4673-0060-5, pp. 12-17, Madrid, Spain, 2011.
- D. Pelusi, "Optimization of a Fuzzy Logic Controller using Genetic Algorithms", proceedings of the IEEE International Conference on Intelligent Human-Machine Systems and Cybernetics, Hangzhou, ISBN: 978-1-4577-0676-9, pp. 143-146, Hangzhou, China, 2011.

- D. Pelusi, "Design and implementation of fuzzy controllers for optimal signals sampling", proceedings of the International Conference on Innovative Computing and Communication and Asia-Pacific Conference on Information Technology and Ocean Engineering, Macao, China, 2011.
- D. Pelusi, LG Tallini, B. Bose, "On m-ary Balanced Codes with Parallel Decoding", Proceedings of the IEEE International Symposium on Information Theory, Austin, Texas, USA, pp. 1305-1309, June 13 - 18, 2010
- D. Pelusi, "A pattern recognition algorithm for optimal profits in currency trading", Mathematical and Statistical Methods for Actuarial Sciences and Finance, Springer, XII, 300 p., Hardcover ISBN: 978-88-470-1480-0, pp. 233-242, 2010
- D. Pelusi, "On finding training and trading time set, for optimal profits in currency trading using Technical Analysis", proceedings of International Conference "Mathematical and Statistical Methods for Actuarial Sciences and Finance 2008", Venice, 2008
- R. Mascella, D. Pelusi, LG Tallini, "On combinatorial algorithms for indexing m-ary word weight classes", Advances in Mathematics of Uncertainty, Performantica, ISBN (10): 973-730-257-5, ISBN (13) : 978-973-730-257-1, pp.71-87, 2006
- R. Mascella, D. Pelusi, LG Tallini, "On computing the indexing function over the set of combinatorial compositions", Advances in Mathematics of Uncertainty, Performantica, ISBN (10): 973-730-257-5, ISBN (13) : 978-973-730-257-1, pp. 59-69, 2006
- F. Eugeni, R. Mascella, D. Pelusi, "Uncertainty from philosophical and mathematical point of view", Advances in Mathematics of Uncertainty, Performantica, ISBN (10): 973-730-257-5, ISBN (13): 978 -973-730-257-1, pp. 7-12, 2006
- F. Bortoletto, M. D'Alessandro, E. Giro, D. Magrin, L. Corcione, D. Pelusi, C. Giuliani, A. Di Cianno, "Control system architecture for AMICA: the Antarctic NIR/MIR camera for IRAIT ", Ground-based and Airborne Instrumentation for Astronomy. Edited by McLean, Ian S.; Iye, Masanori. Proceedings of the SPIE, Volume 6269, pp. 62695A, Orlando, 2006.
- M. Dolci, O. Straniero, G. Valentini, G. Di Rico, M. Ragni, D. Pelusi, I. Di Varano, C. Giuliani, A. Di Cianno, A. Valentini, L. Corcione, F. Bortoletto , M. D'Alessandro, C. Bonoli, E. Giro, D. Fantinel, D. Magrin, F. Zerbi, A. Riva, E. Molinari, P. Conconi, V. De Caprio, M. Busso, G. Tosti, G. Nucciarelli, F. Roncella, C. Abia, "AMICA (Antarctic Multiband Infrared CAmera) Project", Ground-based and Airborne Telescopes. Edited by Stepp, Larry M.. Proceedings of the SPIE, Volume 6267, pp. 62671I, Orlando, 2006.
- F. Bortoletto, M. D'Alessandro, C. Bonoli, D. Fantinel, E. Giro, D. Magrin, L. Corcione, D. Pelusi, C. Giuliani, A. di Cianno, V. De Caprio, A. Riva, "The Antarctic NIR/MIR Camera AMICA", Scientific Detectors for Astronomy 2005. Edited by Jenna E. Beletic, University of Chicago, USA; James W. Beletic, Rockwell Scientific Company, USA; Paola Amico, European Southern Observatory, Chile. Volume LXIV, p. 771 1-4020-4329-5. Berlin: Springer Dordrecht, p.673, 2006
- G. Tosti, M. Busso, G. Nucciarelli, M. Bagaglia , F. Roncella, A. Mancini, S. Castellini , M. Mariotti , E. Babucci , G. Chiocci, O. Straniero, M. Dolci, G. Valentini, I. di Varano, D. Pelusi, G. Di Rico, M. Ragni , C. Abia, I. Domnguez, L. Corcione , F. Porcu , P. Conconi , V. De Caprio , A. Riva , E. Molinari, FM Zerbi, F. Bortoletto, C. Bonoli, M. D'Alessandro, J. Colom, J. Isern, R. Briguglio, A. Cacciani, L. Farnesini, B.

Checucci, KG Strassmeier, "The International Robotic Antarctic Infrared Telescope (IRAIT)", Ground-based and Airborne Telescopes. Edited by Stepp, Larry M.. Proceedings of the SPIE, Volume 6267, pp. 62671H, Orlando, 2006.

- A. Riva, M. Dolci, O. Straniero, FM Zerbi, E. Molinari, P. Conconi, V. De Caprio, G. Valentini, G. Di Rico, M. Ragni, D. Pelusi, I. Di Varano, C. Giuliani, A. Di Cianno, A. Valentini, F. Bortoletto, M. D'Alessandro, C. Bonoli, E. Giro, D. Fantinel, D. Magrin, L. Corcione, M. Busso, G. Tosti, G. Nucciarelli, F. Roncella, C. Abia and the IRAIT Team, "AMICA - The Infrared Eye at DOME C", Astronomy in Antarctica, 26th meeting of the IAU, Prague, 2006
- D. Pelusi, "Fuzzy systems", Journal of Discrete Mathematical Sciences & Cryptography, Volume 8, Nr. 1, pp. 131-136, April 2005
- M. Dolci, G. Valentini, O. Straniero, G Di Rico, M. Ragni, D. Pelusi, C. Giuliani, A. Di Cianno, I. Di Varano, G. Tosti, M. Busso, L. Corcione, "AMICA: the Antarctic Mid-Infrared Camera for the IRAIT telescope", EAS Publications Series, Volume 14, pp. 337-342, 2005
- G. Di Rico, M. Ragni, M. Dolci, G. Valentini, O. Straniero, I. Di Varano, D. Pelusi, D. Magrin, E. Giro, D. Fantinel, C. Bonoli, M. D' Alessandro, F. Bortoletto, L. Corcione, "Infrared Observations from Antarctica: the AMICA Project", proceedings of "Astronomical Data Analysis Software and Systems XV ASP Conference Series", Vol. 351. Edited by Carlos Gabriel, Christophe Arviset, Daniel Ponz , and Enrique Solano. San Francisco: Astronomical Society of the Pacific, p.739, 2006. San Lorenzo de El Escorial, 2005
- G. Tosti, M. Busso, O. Straniero, C. Abia, M Bagaglia, M. Dolci, L. Corcione, G. Nucciarelli, F. Roncella, G. Valentini, I. Di Varano, D. Pelusi, "The IRAIT Project: infrared astronomy from Antarctica", proceedings of SPIE, Volume 5489, pp. 742-753, Glasgow, 2004.
- D. Pelusi, "A fuzzy controller for digital signals processing", Fuzzy systems and AI, EA, Volume 10, Nr. 3, pp. 131-138, 2004.
- G. Di Rico, I. Di Varano, D. Pelusi, M. Ragni, M. Dolci, G. Valentini, O. Straniero, L. Corcione, G. Tosti, M. Busso, "The Antarctic Mid-Ir Camera ( AMICA) for the IRAIT telescope", Astronomische Nachrichten, Volume 325, Issue 6, pp. 664-664, 2004.

## Research products

### Patents:

- V.Vinoth Kumar, G. Sabeena Gnanaselvi, Danilo Pelusi, J. Balajee, IoT-Based Weather Monitoring System Using Embedded Electronics with Wireless Sensor Network, Intellectual Property India. Indian Patent no. 202241039545, July 15, 2022.

- Pelusi D., Cyriac R., Elton RJ, Prabhakar MA, Balamurugan P., Marimuthu K., Malfunction of Hard Disk Drive Recognition and Predictive Analysis Using AI and MI Techniques. Intellectual Property Australia. Australian Patent no. 2021104777, July 31, 2021.

- Bradha M, Pelusi D., Nagaraj B, Design and development of exible one-dimensional hybrid SmCrO<sub>3</sub>-CNT perovskite-based nanosensor for explosive trace detection. Intellectual Property India. Indian Patent no. 202141019856, April 29, 2021.